

Course Description

1: General Information

Course Name: Data and systems Security Course Code: ITNM3319
Level of Course: 3
Credits – 3 Lecture: 2 Laboratory: 2 Tutorial: 0
Prerequisites: Computer Networks , Operating Systems.
Year / Semester: 2 nd semester 2025/2026
Lecturer: Eng. Hassan M. Dawoud

2: Course description

This course provides students with a balanced understanding of the technical and organizational issues related to Information Security. Students will receive theoretical and practical instructions in both technical and managerial aspects of securing information in organizations. Representative topics covered in the course include key topics such as Internet and network security, Encryption, Cryptography, Symmetric and Public Key algorithms, Digital signatures, Authentication, Key database security, secure communications and industry best practices for information assurance. The course will be helpful to the students who aim to pursue Certified Information Systems Security Professional certification and/or careers in Information Security.

3: Course aims and Intended Learning Outcomes-

A- Aims:

The main goal of this course is to prepare students to become able to understand and implement the principles of security and protection of systems including:

Topics include:

1. Introduction to Information Security
2. Security Threats 1
3. Network Attacks
4. Firewall
5. Wireless and mobile Network Security
6. Cybersecurity Resilience and Privacy
7. Basic Cryptography
8. Public key Infrastructure and cryptographic protocol

4: Semester Teaching plan (please write in details the topics to be covered)

Lecture No.	Topics	Objectives	Week
1.	Course admission		1
2.	Introduction to Information Security	1. Information Security Concepts 2. Security Requirements 3. Security Challenges 4. Security Attacks 5. Security Services 6. Security Models	2
3.	Security Threats	1. Information Security Terminologies 2. Difficulties in Defending Against Attacks 3. Considerations Defending Against Attacks 4. Malicious software (Malware) 5. Psychological Approaches	3
4.	Network Attacks	1. Networking-Based Attacks 2. Server-Based Attacks	4
5.	Firewall	1. Firewall 2. Virtual Private Network Concentrator 3. Mail Gateway 4. Intrusion Detection System (IDS) 5. Intrusion Prevention System (IPS)	5
6.	Wireless Network Security	1. Bluetooth Systems 2. Near Field Communication Systems 3. Radio Frequency Identification Systems 4. Wi-Fi Systems	6

		5. Security Solutions	
8. Midterm Exam			8
7.	Introduction to Cryptography	1. Introduction 2. Cryptographic Transformations 3. Stream and Block Cipher 4. Feistel cipher 5. Data Encryption Standard (DES)	10
8.	Symmetric Cryptography	1. Advanced Encryption Standard 2. Double and Triple DES 3. Modes of Operation	11
9.	Asymmetric Cryptography	1. Introduction 2. RSA Algorithm 3. Elgamal Algorithm 4. Diffie-Hellman Algorithm	13
15. Final Exam			15

5: Evaluation tasks (it's only a sample)

No	Task	Marks
1	Practical	30
2	Midterm Exam	20
3	Final Exam	50